

GEOG 347 Assignment 4

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```
#install.packages("ISLR2")
```

```
library(ISLR2)  
data(Auto)
```

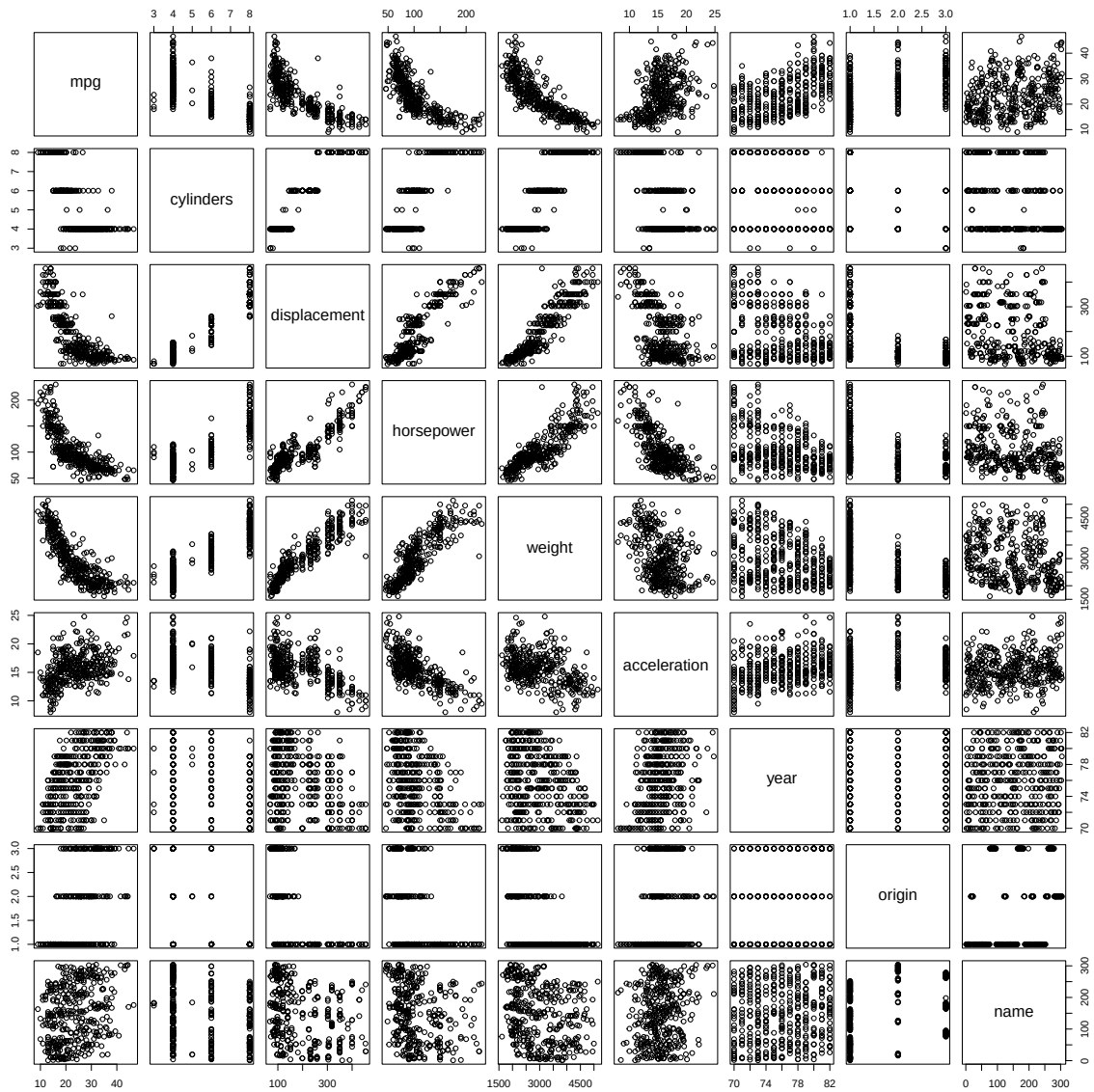
Part 1

1.a

```
View(Auto)  
?Auto
```

```
starting httpd help server ... done
```

```
auto_vars <- Auto[, c("mpg", "cylinders", "displacement", "horsepower", "weight", "acceleration", "disposition")]  
pairs(auto_vars)
```



1.b

```
auto_vars <- Auto[, c("mpg", "cylinders", "displacement", "horsepower", "weight", "accelerat.  
df = subset(auto_vars, select = -c(name) )  
cor(df)
```

mpg cylinders displacement horsepower weight

mpg	1.0000000	-0.7776175	-0.8051269	-0.7784268	-0.8322442
cylinders	-0.7776175	1.0000000	0.9508233	0.8429834	0.8975273
displacement	-0.8051269	0.9508233	1.0000000	0.8972570	0.9329944
horsepower	-0.7784268	0.8429834	0.8972570	1.0000000	0.8645377
weight	-0.8322442	0.8975273	0.9329944	0.8645377	1.0000000
acceleration	0.4233285	-0.5046834	-0.5438005	-0.6891955	-0.4168392
year	0.5805410	-0.3456474	-0.3698552	-0.4163615	-0.3091199
origin	0.5652088	-0.5689316	-0.6145351	-0.4551715	-0.5850054

	acceleration	year	origin
mpg	0.4233285	0.5805410	0.5652088
cylinders	-0.5046834	-0.3456474	-0.5689316
displacement	-0.5438005	-0.3698552	-0.6145351
horsepower	-0.6891955	-0.4163615	-0.4551715
weight	-0.4168392	-0.3091199	-0.5850054
acceleration	1.0000000	0.2903161	0.2127458
year	0.2903161	1.0000000	0.1815277
origin	0.2127458	0.1815277	1.0000000

According to these result, cylinders and displacement have the strongest correlation with 0.9508233, followed by weight and displacement. The weakest correlation appears to be between weight and mpg with -0.8322442.

1.c

```
lm.fit <- lm(mpg ~ cylinders+displacement+horsepower+weight+acceleration+year+origin, data =
summary(lm.fit))
```

Call:

```
lm(formula = mpg ~ cylinders + displacement + horsepower + weight +
    acceleration + year + origin, data = Auto)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-9.5903	-2.1565	-0.1169	1.8690	13.0604

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-17.218435	4.644294	-3.707	0.00024	***
cylinders	-0.493376	0.323282	-1.526	0.12780	
displacement	0.019896	0.007515	2.647	0.00844	**

```

horsepower    -0.016951    0.013787   -1.230   0.21963
weight        -0.006474    0.000652   -9.929   < 2e-16 ***
acceleration   0.080576    0.098845    0.815   0.41548
year           0.750773    0.050973   14.729   < 2e-16 ***
origin         1.426141    0.278136    5.127   4.67e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 3.328 on 384 degrees of freedom
Multiple R-squared:  0.8215,    Adjusted R-squared:  0.8182
F-statistic: 252.4 on 7 and 384 DF,  p-value: < 2.2e-16

```

The intercept value is -17.218435. The predictors all have slopes that positively or negatively align with the values from correlation table, except for displacement. Displacement has a positive correlation but a negative slope in the linear model function. The R squared value is 0.8215 which means that 82.15% of variation in y is explained by x.

1.c.i

The F-statistic (252.4 on 7 and 384 DF) and p-value (p-value: < 2.2e-16) both indicate that the joint null hypothesis (all the regression coefficients, excluding the intercept, are equal to zero) would be rejected. It is clear from the summary table that some of the predictors have a non-zero impact on y. The p-value is far below an assumed $\alpha=0.05$ there is an extremely low chance of falsely rejecting the null hypothesis (type 1 error).

1.c.ii

Displacement, weight, year, and origin all appear to have a statistically significant relationship with the response variable. Each of these predictors have a reported p-value of far below the assumed $\alpha=0.05$. This means that the values of these predictors are extremely unlikely due to random chance.

1.c.iii

The coefficient for year is 0.750773, which suggests there is a positive linear relationship (slope) between year and mpg. That means for every unit of year increase there is a 0.750773 increase in the mpg value.

1.d

```
#fit all is the backward selection method, starting with all variables and removing for best
fit_all <- lm(mpg ~ . -name, data = Auto)
back_sel <- step(fit_all, direction = 'backward')
```

Start: AIC=950.5

```
mpg ~ (cylinders + displacement + horsepower + weight + acceleration +
      year + origin + name) - name
```

	Df	Sum of Sq	RSS	AIC
- acceleration	1	7.36	4259.6	949.18
- horsepower	1	16.74	4269.0	950.04
<none>			4252.2	950.50
- cylinders	1	25.79	4278.0	950.87
- displacement	1	77.61	4329.8	955.59
- origin	1	291.13	4543.3	974.46
- weight	1	1091.63	5343.8	1038.08
- year	1	2402.25	6654.5	1124.06

Step: AIC=949.18

```
mpg ~ cylinders + displacement + horsepower + weight + year +
      origin
```

	Df	Sum of Sq	RSS	AIC
<none>			4259.6	949.18
- cylinders	1	27.27	4286.8	949.68
- horsepower	1	53.80	4313.4	952.10
- displacement	1	73.57	4333.1	953.89
- origin	1	292.02	4551.6	973.17
- weight	1	1310.43	5570.0	1052.32
- year	1	2396.17	6655.7	1122.13

```
summary(back_sel)
```

Call:

```
lm(formula = mpg ~ cylinders + displacement + horsepower + weight +
    year + origin, data = Auto)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.7604	-2.1791	-0.1535	1.8524	13.1209

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1.556e+01	4.175e+00	-3.728	0.000222	***
cylinders	-5.067e-01	3.227e-01	-1.570	0.117236	
displacement	1.927e-02	7.472e-03	2.579	0.010287	*
horsepower	-2.389e-02	1.084e-02	-2.205	0.028031	*
weight	-6.218e-03	5.714e-04	-10.883	< 2e-16	***
year	7.475e-01	5.079e-02	14.717	< 2e-16	***
origin	1.428e+00	2.780e-01	5.138	4.43e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.326 on 385 degrees of freedom

Multiple R-squared: 0.8212, Adjusted R-squared: 0.8184

F-statistic: 294.6 on 6 and 385 DF, p-value: < 2.2e-16

#fit null is the forward selection method, starting with ~1, or only intercept, and adding m

```
fit_null <- lm(mpg ~ 1, data = Auto)
forward_sel <- step(fit_null,
                    scope = list(lower = fit_null, upper = fit_all),
                    direction = 'forward')
```

Start: AIC=1611.93

mpg ~ 1

	Df	Sum of Sq	RSS	AIC
+ weight	1	16497.8	7321.2	1151.5
+ displacement	1	15440.2	8378.8	1204.4
+ horsepower	1	14433.1	9385.9	1248.9
+ cylinders	1	14403.1	9415.9	1250.1
+ year	1	8027.7	15791.3	1452.8
+ origin	1	7609.2	16209.8	1463.1
+ acceleration	1	4268.5	19550.5	1536.5
<none>			23819.0	1611.9

Step: AIC=1151.49

mpg ~ weight

	Df	Sum of Sq	RSS	AIC
+ year	1	2752.28	4569.0	968.66
+ horsepower	1	327.39	6993.8	1135.56

```

+ origin      1    222.25 7099.0 1141.41
+ acceleration 1    168.34 7152.9 1144.37
+ displacement 1    150.93 7170.3 1145.33
+ cylinders    1    115.12 7206.1 1147.28
<none>                7321.2 1151.49

```

Step: AIC=968.66
mpg ~ weight + year

	Df	Sum of Sq	RSS	AIC
+ origin	1	220.847	4348.1	951.24
<none>			4569.0	968.66
+ acceleration	1	10.450	4558.5	969.77
+ cylinders	1	4.958	4564.0	970.24
+ horsepower	1	3.302	4565.7	970.38
+ displacement	1	0.042	4568.9	970.66

Step: AIC=951.24
mpg ~ weight + year + origin

	Df	Sum of Sq	RSS	AIC
<none>			4348.1	951.24
+ displacement	1	15.3765	4332.7	951.85
+ acceleration	1	15.0322	4333.1	951.89
+ horsepower	1	14.4048	4333.7	951.94
+ cylinders	1	0.1476	4348.0	953.23

```
summary(forward_sel)
```

Call:
lm(formula = mpg ~ weight + year + origin, data = Auto)

Residuals:

Min	1Q	Median	3Q	Max
-9.9440	-2.0948	-0.0389	1.7255	13.2722

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.805e+01	4.001e+00	-4.510	8.60e-06 ***
weight	-5.994e-03	2.541e-04	-23.588	< 2e-16 ***
year	7.571e-01	4.832e-02	15.668	< 2e-16 ***

```
origin          1.150e+00  2.591e-01  4.439 1.18e-05 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 3.348 on 388 degrees of freedom
```

```
Multiple R-squared:  0.8175,    Adjusted R-squared:  0.816
```

```
F-statistic: 579.2 on 3 and 388 DF,  p-value: < 2.2e-16
```

```
#mixed selection method will start with fit null lm and add/remove variables as needed to re
```

```
mixed_sel <- step(fit_null,  
                  scope = list(lower = fit_null, upper = fit_all),  
                  direction = 'both')
```

```
Start:  AIC=1611.93
```

```
mpg ~ 1
```

	Df	Sum of Sq	RSS	AIC
+ weight	1	16497.8	7321.2	1151.5
+ displacement	1	15440.2	8378.8	1204.4
+ horsepower	1	14433.1	9385.9	1248.9
+ cylinders	1	14403.1	9415.9	1250.1
+ year	1	8027.7	15791.3	1452.8
+ origin	1	7609.2	16209.8	1463.1
+ acceleration	1	4268.5	19550.5	1536.5
<none>			23819.0	1611.9

```
Step:  AIC=1151.49
```

```
mpg ~ weight
```

	Df	Sum of Sq	RSS	AIC
+ year	1	2752.3	4569.0	968.66
+ horsepower	1	327.4	6993.8	1135.56
+ origin	1	222.2	7099.0	1141.41
+ acceleration	1	168.3	7152.9	1144.37
+ displacement	1	150.9	7170.3	1145.33
+ cylinders	1	115.1	7206.1	1147.28
<none>			7321.2	1151.49
- weight	1	16497.8	23819.0	1611.93

```
Step:  AIC=968.66
```

```
mpg ~ weight + year
```


	Df	Sum of Sq	RSS	AIC
+ origin	1	220.8	4348.1	951.24
<none>			4569.0	968.66
+ acceleration	1	10.5	4558.5	969.77
+ cylinders	1	5.0	4564.0	970.24
+ horsepower	1	3.3	4565.7	970.38
+ displacement	1	0.0	4568.9	970.66
- year	1	2752.3	7321.2	1151.49
- weight	1	11222.4	15791.3	1452.81

Step: AIC=951.24

mpg ~ weight + year + origin

	Df	Sum of Sq	RSS	AIC
<none>			4348.1	951.24
+ displacement	1	15.4	4332.7	951.85
+ acceleration	1	15.0	4333.1	951.89
+ horsepower	1	14.4	4333.7	951.94
+ cylinders	1	0.1	4348.0	953.23
- origin	1	220.8	4569.0	968.66
- year	1	2750.9	7099.0	1141.41
- weight	1	6235.4	10583.5	1297.95

```
summary(mixed_sel)
```

Call:

```
lm(formula = mpg ~ weight + year + origin, data = Auto)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.9440	-2.0948	-0.0389	1.7255	13.2722

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.805e+01	4.001e+00	-4.510	8.60e-06 ***
weight	-5.994e-03	2.541e-04	-23.588	< 2e-16 ***
year	7.571e-01	4.832e-02	15.668	< 2e-16 ***
origin	1.150e+00	2.591e-01	4.439	1.18e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.348 on 388 degrees of freedom
 Multiple R-squared: 0.8175, Adjusted R-squared: 0.816
 F-statistic: 579.2 on 3 and 388 DF, p-value: < 2.2e-16

The selected models are the same for mixed and forward selection with AIC=951.24 when $\text{mpg} \sim \text{weight} + \text{year} + \text{origin}$. The F statistic is also the same for these two selected models at 579.2 on 3 and 388 DF. The residual values are also the same for forward and mixed selection methods. However, the backward selection method produces a better AIC and contains more variables with AIC=949.18 when $\text{mpg} \sim \text{cylinders} + \text{displacement} + \text{horsepower} + \text{weight} + \text{year} + \text{origin}$. The F statistic for this model is F-statistic: 294.6 on 6 and 385 DF. The residual values are also different for this selection method than forward and mixed methods.

1.e

```
#was unable to get this result when using the code: fit_interact <- lm(mpg ~ . -name^2, data = Auto)
fit_interact <- lm(mpg ~ (cylinders+displacement+horsepower+weight+acceleration+year+origin))
summary(fit_interact)
```

Call:

```
lm(formula = mpg ~ (cylinders + displacement + horsepower + weight +
  acceleration + year + origin)^2, data = Auto)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.6303	-1.4481	0.0596	1.2739	11.1386

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.548e+01	5.314e+01	0.668	0.50475
cylinders	6.989e+00	8.248e+00	0.847	0.39738
displacement	-4.785e-01	1.894e-01	-2.527	0.01192 *
horsepower	5.034e-01	3.470e-01	1.451	0.14769
weight	4.133e-03	1.759e-02	0.235	0.81442
acceleration	-5.859e+00	2.174e+00	-2.696	0.00735 **
year	6.974e-01	6.097e-01	1.144	0.25340
origin	-2.090e+01	7.097e+00	-2.944	0.00345 **
cylinders:displacement	-3.383e-03	6.455e-03	-0.524	0.60051
cylinders:horsepower	1.161e-02	2.420e-02	0.480	0.63157
cylinders:weight	3.575e-04	8.955e-04	0.399	0.69000
cylinders:acceleration	2.779e-01	1.664e-01	1.670	0.09584 .

cylinders:year	-1.741e-01	9.714e-02	-1.793	0.07389	.
cylinders:origin	4.022e-01	4.926e-01	0.816	0.41482	
displacement:horsepower	-8.491e-05	2.885e-04	-0.294	0.76867	
displacement:weight	2.472e-05	1.470e-05	1.682	0.09342	.
displacement:acceleration	-3.479e-03	3.342e-03	-1.041	0.29853	
displacement:year	5.934e-03	2.391e-03	2.482	0.01352	*
displacement:origin	2.398e-02	1.947e-02	1.232	0.21875	
horsepower:weight	-1.968e-05	2.924e-05	-0.673	0.50124	
horsepower:acceleration	-7.213e-03	3.719e-03	-1.939	0.05325	.
horsepower:year	-5.838e-03	3.938e-03	-1.482	0.13916	
horsepower:origin	2.233e-03	2.930e-02	0.076	0.93931	
weight:acceleration	2.346e-04	2.289e-04	1.025	0.30596	
weight:year	-2.245e-04	2.127e-04	-1.056	0.29182	
weight:origin	-5.789e-04	1.591e-03	-0.364	0.71623	
acceleration:year	5.562e-02	2.558e-02	2.174	0.03033	*
acceleration:origin	4.583e-01	1.567e-01	2.926	0.00365	**
year:origin	1.393e-01	7.399e-02	1.882	0.06062	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.695 on 363 degrees of freedom

Multiple R-squared: 0.8893, Adjusted R-squared: 0.8808

F-statistic: 104.2 on 28 and 363 DF, p-value: < 2.2e-16

According to this summary, it appears like the interaction between acceleration and origin are the most statistically significant. There is also a positive direction of this coefficient, meaning that as the origin unit increases, there is a positive affect of the acceleration variable. Acceleration and year, and displacement and year also appear to be statistically significant as well (along with positive direction of coefficient).

Part 2

```
data(Carseats)
View(Carseats)
?Carseats
```

2.a

```
fit_carseats <- lm(Sales ~ Price + Urban + US, data = Carseats)
summary(fit_carseats)
```

Call:

```
lm(formula = Sales ~ Price + Urban + US, data = Carseats)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.9206	-1.6220	-0.0564	1.5786	7.0581

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.043469	0.651012	20.036	< 2e-16 ***
Price	-0.054459	0.005242	-10.389	< 2e-16 ***
UrbanYes	-0.021916	0.271650	-0.081	0.936
USYes	1.200573	0.259042	4.635	4.86e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.472 on 396 degrees of freedom

Multiple R-squared: 0.2393, Adjusted R-squared: 0.2335

F-statistic: 41.52 on 3 and 396 DF, p-value: < 2.2e-16

2.b

There is a negative linear relationship between Price and Sales; for every unit increase of price, there is a -0.054459 coefficient on the sales unit. Urban and US fields are both qualitative; in this case, the data is either a Yes or No value. The coefficients in this summary for Urban and US are also measuring the Yes value. So for Urban Yes, the coefficient is -0.021916, which means that for every x unit increase (more urban predictor units), there is a negative linear relationship with sales by -0.021916. For US Yes, the coefficient is 1.200573, which means that for every x unit increase (more US predictor units) there is a positive linear relationship with sales.

2.c

$$\text{Sales} = \beta_0 + \beta_1 \text{Price} + \beta_2 \text{UrbanYES} + \beta_3 \text{USYES} + \epsilon$$

2.d

According to the summary table above, I see that Price and USYes $\Pr(>|t|)$ values are statistically significant; the null hypothesis can be rejected for these two variables. The F statistic value, along with the p-value for the whole model also indicate that at least one of the variables variables can reject the null hypothesis, further validating rejection for Price and USYes.

2.e

```
confint(fit_carseats)
```

		2.5 %	97.5 %
(Intercept)	11.76359670	14.32334118	
Price	-0.06476419	-0.04415351	
UrbanYes	-0.55597316	0.51214085	
USYes	0.69130419	1.70984121	